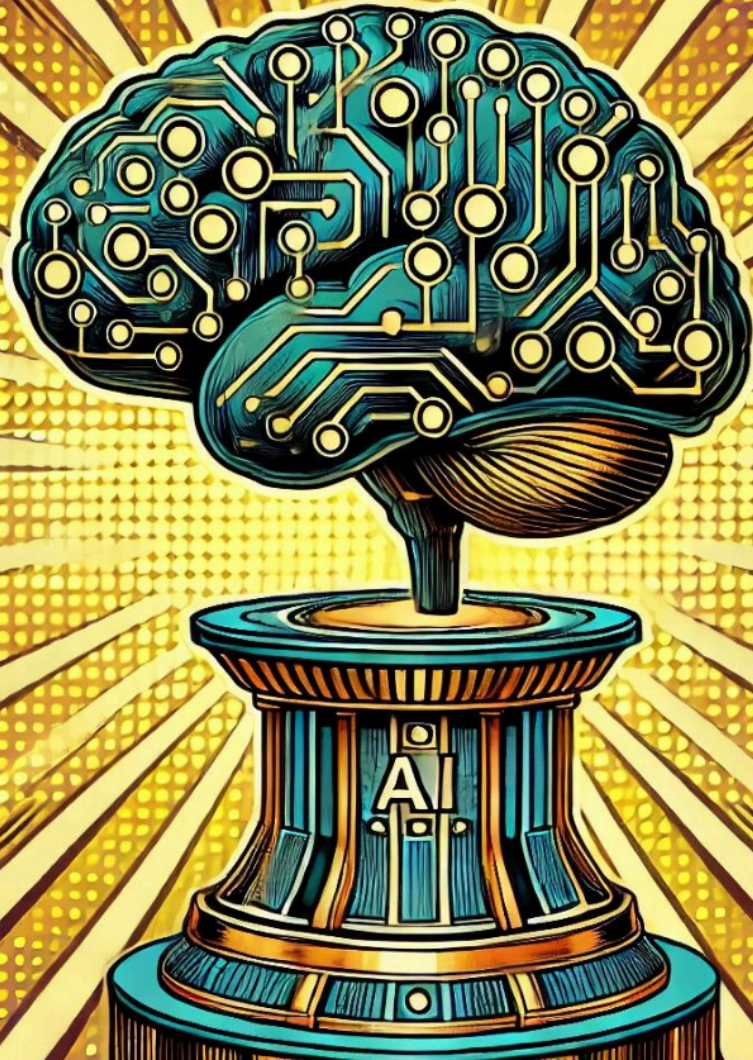


LLM Engineering

MASTER AI & LARGE LANGUAGE MODELS





WEEK 8

GAME ON

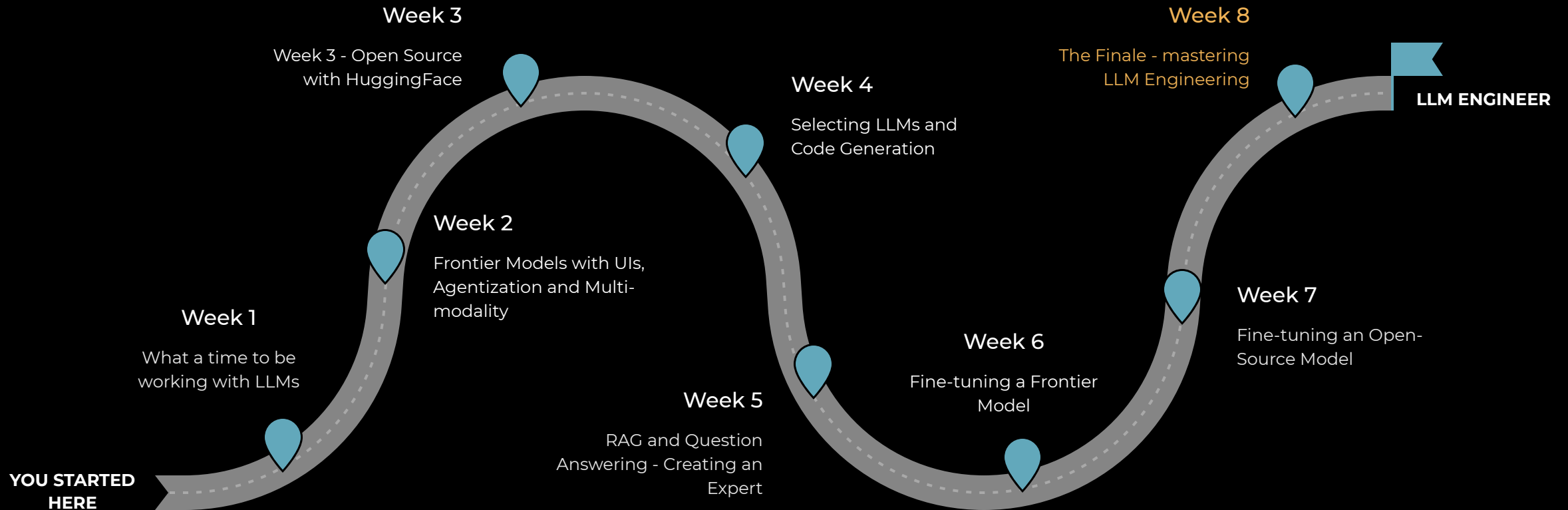
What you can now do

- Generate text and code with Frontier Models and Open Source models using APIs and HuggingFace, including tools, assistants and RAG
- Follow a 5 step strategy to solve problems, including dataset curation, making a baseline model, and fine-tuning a Frontier model
- Confidently carry out the end-to-end process for selecting and training specialized open source models that can outperform the Frontier

An intensive week lies ahead! After today, you'll be able to:

- Use Modal, the serverless platform for AI, to run code remotely
- Deploy an LLM behind an API in the cloud
- Create an Agent that will be incorporated into an autonomous Agentic AI solution

This week the training wheels are OFF



It's all been building up to this

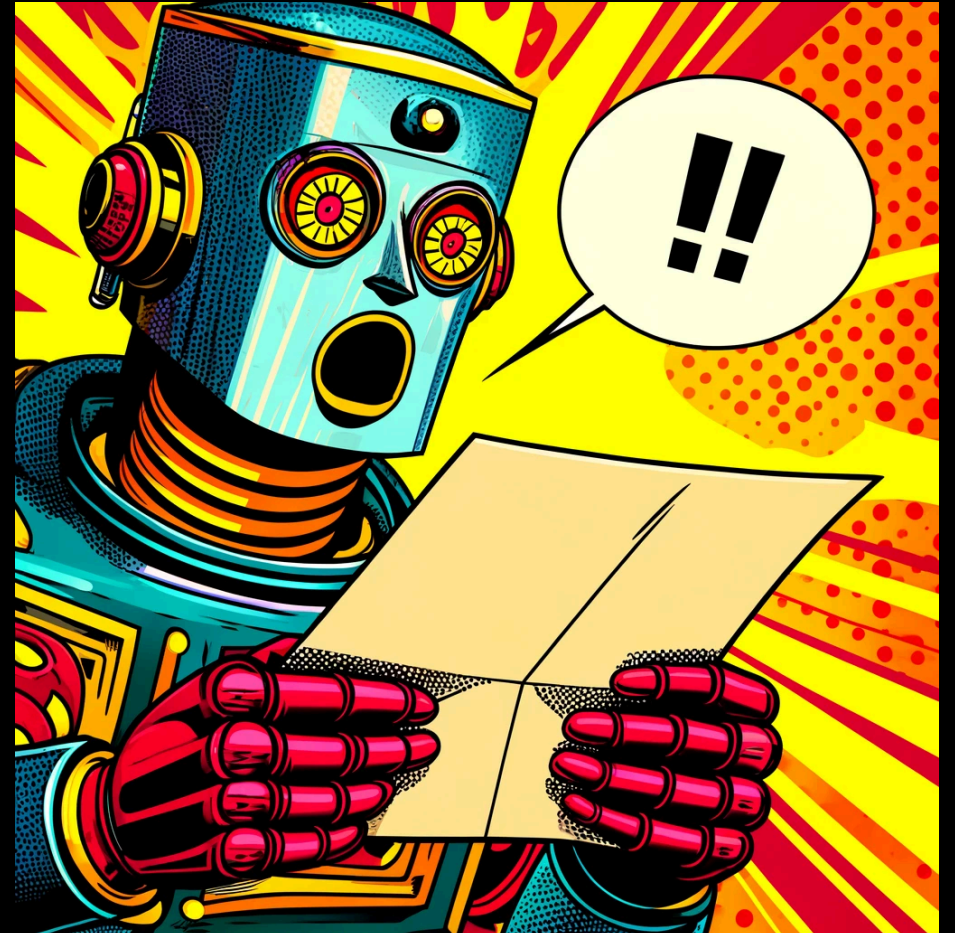
Capstone Project - "The Price Is Right"

We will build an Autonomous Agentic AI framework

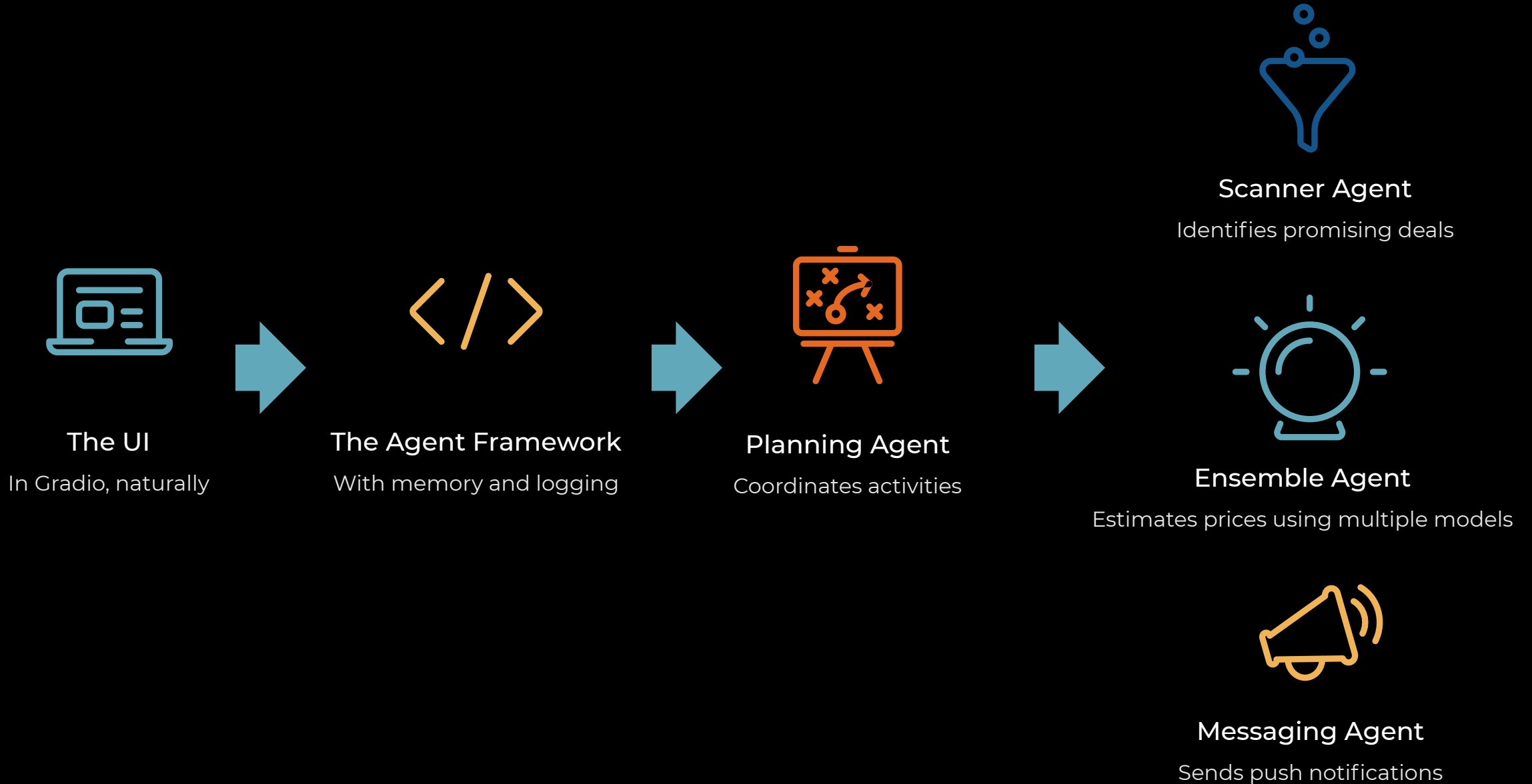
- Watches for deals being published online
- Estimates the price of the product
- Sends a push notification when it finds great opportunities

SEVEN Agents will collaborate in our framework

- GPT-4o model will identify the deals from an RSS feed
- Our frontier-busting fine-tuned model will estimate prices
- We'll also use a Frontier Model with a massive RAG db



Agent Architecture



As we head towards productionizing



Type hints

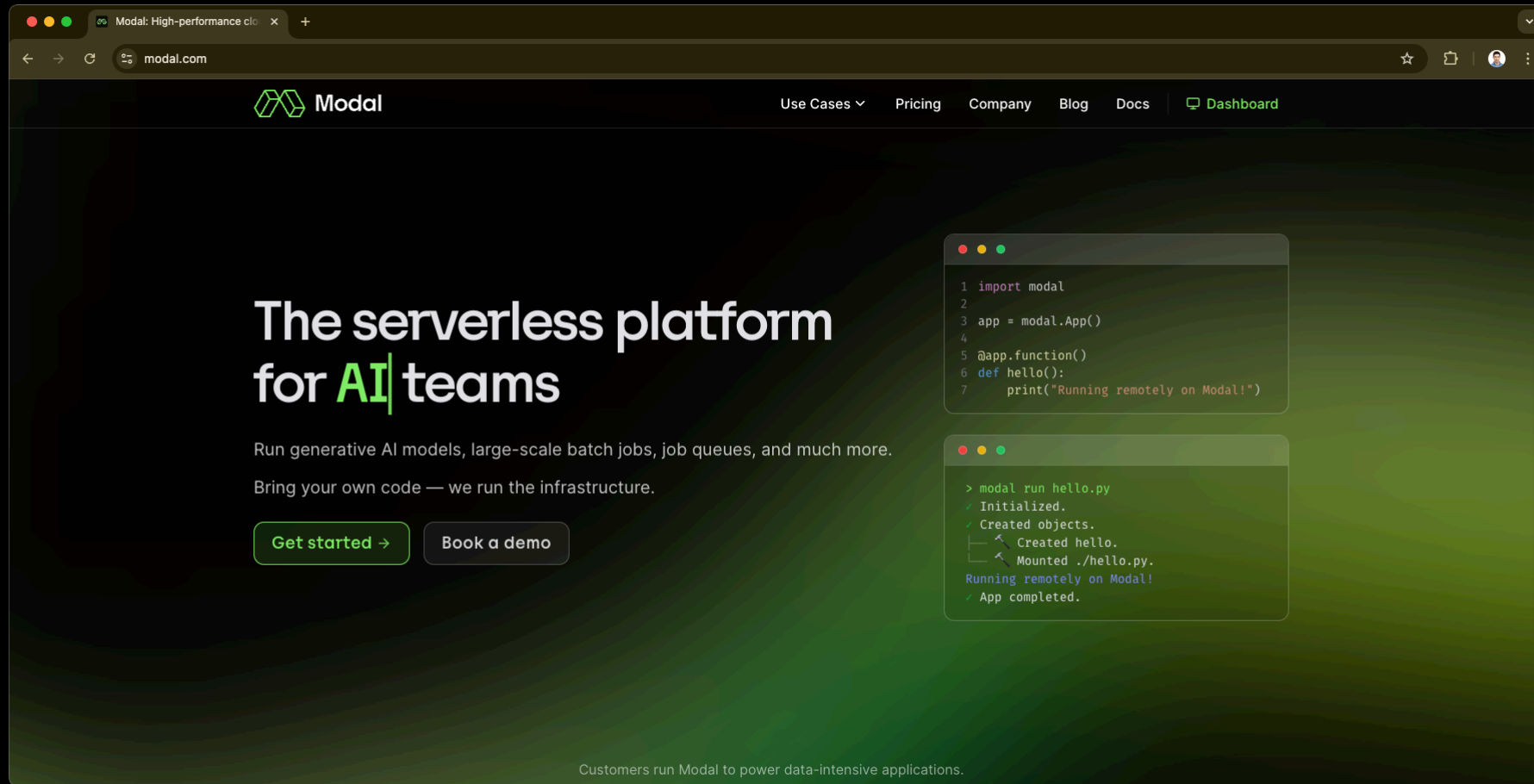


Comments



Logging

Introducing Modal



The screenshot shows the Modal website homepage in a web browser. The browser's address bar shows 'modal.com'. The website has a dark green background with a subtle pattern. The Modal logo is in the top left, and navigation links for 'Use Cases', 'Pricing', 'Company', 'Blog', 'Docs', and 'Dashboard' are in the top right. The main heading reads 'The serverless platform for AI teams', with 'AI' in a light green font. Below this, text describes the platform's capabilities: 'Run generative AI models, large-scale batch jobs, job queues, and much more. Bring your own code — we run the infrastructure.' Two buttons, 'Get started →' and 'Book a demo', are positioned below the text. To the right, two code snippets are displayed in light green boxes. The first snippet shows Python code for creating a Modal app and a function. The second snippet shows the terminal output of running the code, indicating successful initialization and execution on Modal.

Modal

Use Cases ▾ Pricing Company Blog Docs Dashboard

The serverless platform for AI teams

Run generative AI models, large-scale batch jobs, job queues, and much more.
Bring your own code — we run the infrastructure.

[Get started →](#) [Book a demo](#)

```
1 import modal
2
3 app = modal.App()
4
5 @app.function()
6 def hello():
7     print("Running remotely on Modal!")
```

```
> modal run hello.py
✓ Initialized.
✓ Created objects.
  ↳ Created hello.
  ↳ Mounted ./hello.py.
Running remotely on Modal!
✓ App completed.
```

Customers run Modal to power data-intensive applications.



WEEK 8

I told you this week would be big!

What you can now do

- Generate text and code with Frontier Models and Open Source models using APIs and HuggingFace, including tools, assistants and RAG
- Follow a 5 step strategy to solve problems, including dataset curation, making a baseline model, and fine-tuning a Frontier model
- Confidently carry out the end-to-end process for selecting and training specialized open source models that can outperform the Frontier
- Use Modal to deploy LLMs for production

By end of next time you'll be able to:

- Confidently build an advanced RAG solution, without Langchain
- Build an ensemble model with a high level of expertise
- Deliver production ready code to call a number of models